

Prognostic Markers for Poor Recovery After Mild Traumatic Brain Injury in Older Adults: A Pilot Cohort Study

Vicki L. Kristman, PhD; Robert J. Brison, MD, MPH, FRCP(C), CCFP; Michel Bédard, PhD; Paula Reguly, MPH; Shelley Chisholm, BScN

Objective: To identify prognostic markers associated with poor recovery from mild traumatic brain injury (MTBI) in older adults. **Setting:** Three Ontario emergency departments. **Participants:** Forty-nine participants aged 65 years and older that visited an emergency department for MTBI. **Design:** Pilot prospective cohort study. **Main Measures:** Recovery from MTBI determined using the Rivermead Postconcussion symptom Questionnaire, the Glasgow Outcomes Scale–Extended, physical and mental health functioning (SF-12), and a single question on self-rated recovery assessed by telephone shortly after emergency department visit (baseline) and again 6 months later. Predictors were measured at baseline. **Results:** Markers potentially associated with poor recovery included reporting worse health 1 year before the injury, poor expectations for recovery, depression, and fatigue. **Conclusion:** Recovery after MTBI in older adults may be associated more with psychosocial than with biomedical or injury-related factors. **Key words:** adult, aged, brain injuries, depression, emergency service, fatigue, follow-up studies, hospital, prognosis, prospective studies

MILD TRAUMATIC BRAIN INJURY (MTBI, also known as “concussion”) is a common injury in the general population, accounting for 70% to 90% of all traumatic brain injuries and estimated to occur in 600 per 100 000 adults each year.¹ Prognosis is generally good, with most patients recovering within 3 months,² but self-reported symptoms can be persistent in the longer term.³

Author Affiliations: Department of Health Sciences, Lakehead University, Thunder Bay, Ontario, Canada (Drs Kristman and Bédard and Ms Reguly); Division of Human Sciences, Northern Ontario School of Medicine, Lakehead University, Thunder Bay, Ontario, Canada (Drs Kristman and Bédard); Centre for Education and Research on Aging & Health, Lakehead University, Thunder Bay, Ontario, Canada (Drs Kristman and Bédard); Dalla Lana School of Public Health, University of Toronto, Toronto, Ontario, Canada (Dr Kristman); Department of Emergency Medicine, Queen’s University, Kingston General Hospital, Kingston, Ontario, Canada (Dr Brison); St. Joseph’s Care Group, Thunder Bay, Ontario, Canada (Dr Bédard); and Thunder Bay Regional Health Sciences Centre, Thunder Bay, Ontario, Canada (Ms Chisholm).

This project was funded by the Ontario Neurotrauma Foundation, grant reference number 2012-ABI-MTBI-EP-950. Dr Kristman is supported by a CIHR New Investigator Award in Community-Based Primary Health Care.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal’s Web site (www.headtraumarehab.com).

The authors declare no conflicts of interest.

Corresponding Author: Vicki L. Kristman, PhD, Department of Health Sciences, Lakehead University, 955 Oliver Rd, Thunder Bay, Ontario, Canada P7B 5E1 (vkristman@lakeheadu.ca).

DOI: 10.1097/HTR.0000000000000226

Mild traumatic brain injury is also a significant public health problem for older adults; they have the second highest rate of MTBI (following youth and young adults).^{1,4} As the world’s population continues to age, the number of older adults sustaining a brain injury will increase. However, few studies address the prognosis of MTBI in this age group.

Might prognostic markers differ for older adults? With aging, there are significant declines in strength, cardiovascular fitness, and energy level^{5,6} that may influence the number of symptoms after an MTBI as well as the ability of older adults to recover physically from an MTBI. Older individuals also have higher rates of chronic conditions and neurological diseases. Thompson et al⁷ found that 73% of older adults had a medical condition before they sustained a traumatic brain injury compared with 23% of younger adults. Comorbidities and the medications that older adults take for these conditions may influence the course of recovery for older adults. Older adults also have higher rates of depression⁸ and suicide,⁹ potentially leading to reduced recovery and increased mortality after an MTBI. Income levels generally decline and the living environment may change as older adults move out of the workforce and into retirement. All of these markers could influence the ability of older adults to recover after an MTBI.

Clinical prediction rules for prognosis might assist clinicians in advising elderly patients about expectations for recovery. Only 1 clinical prediction rule for MTBI

was identified in a recent systematic review,³ and this study excluded patients older than 60 years.¹⁰ That study found that an absence of comorbid physical problems as well as low levels of both early posttraumatic symptoms and early posttraumatic stress predicted a good outcome at 6 months post-MTBI with a probability of 90%.¹⁰

In 2004, the World Health Organization collaborating center task force on MTBI highlighted the need to assess the effect of older age on recovery from MTBI.² A few case series and descriptive studies suggested that elderly people have poorer recovery after MTBI,¹¹⁻¹³ but no etiologic studies had addressed prognostic factors.² More recently, the International Collaboration on Mild Traumatic Brain Injury Prognosis published a series of articles on prognosis after MTBI.^{3,14-25} The systematic reviews suggested that psychosocial rather than biomedical or injury factors best predicted self-reported recovery in younger adults.^{3,26} This best evidence synthesis again found a lack of published articles assessing MTBI prognosis in the elderly. Urgent priority research areas identified in the synthesis specifically emphasized the need for studies of prognosis after MTBI in the elderly and the development of clinical prediction rules for MTBI to assist clinicians.²⁵ To address these gaps, we initiated a research program with the long-term goal of developing a clinical prediction tool for prognosis of MTBI in older adults. The first step was the current pilot cohort study of prognosis in the older adult population. The primary aim was to identify variables possibly associated with poor recovery after MTBI sustained by older adults. Therefore, according to prognostic study definitions identified in the study by Salmi et al,¹⁴ the goal is to determine noncausal associations (prognostic markers) in a phase I exploratory study. We hypothesized that similar to younger adults, psychosocial markers, rather than biomedical or injury markers, would be identified as having prognostic value for older adults.

METHODS

Participants

Participants included older adults (aged ≥ 65 years) visiting the emergency department (ED) of 1 of 3 hospitals (in Thunder Bay and Kingston, Ontario, Canada) for an MTBI. Participants were invited to participate if they were at least 65 years of age, Ontario residents, and seen in the ED for MTBI within 72 hours of injury. We excluded individuals who did not speak English since we did not have valid, culturally adapted versions of the instruments used in the study. We also excluded potential participants who were unable to provide information because of poor health or incompetency at the time of consent, the baseline data collection phase, or at the 6-month follow-up. All older adult patients meeting the inclusion/exclusion criteria between November

5, 2012, and November 13, 2013, were invited to participate. All participants were contacted by telephone within 10 days of their ED visit for invitation to the study and baseline computer-assisted telephone interview, except for 1 who was reached 16 days after the ED visit. The project was approved by the Lakehead University Research Ethics Board (project # 035 12-13), Thunder Bay Regional Health Sciences Centre (project # 2012116), and Queen's University Health Sciences Research Ethics Board (project #EMED-181-12).

Study design

We used a single-group prospective inception cohort study design. Participants completed a computer-assisted telephone interview survey at baseline. Baseline preinjury and injury data collected included (1) sociodemographic characteristics; (2) psychological and mental health factors; (3) preinjury functional status and brain health; (4) injury-related factors such as mechanism of injury, other injuries sustained, and so forth; (5) comorbidities; and (6) previous brain injuries. In addition, we asked for consent to access information from the ED chart. Six months after the ED visit, participants who had completed the baseline questionnaire were contacted to complete another computer-assisted telephone interview to determine their recovery status. Participants were contacted at 6 months post-ED visit regardless of their recovery status. During the follow-up interviews, participants provided self-assessed recovery information. We also requested information to update some time-dependent variables collected at baseline such as healthcare providers consulted, pain intensity, depression, posttraumatic stress, fatigue, and coping.

We provided the consent form to potential participants in person at the time of the ED visit or by mail. Follow-up was by telephone to complete the interview within 10 days of injury. We made 5 attempts to contact participants by phone at varying times of the day for each planned interview. Those we could not reach after 5 attempts were considered nonrespondents.

For those who participated in the baseline questionnaire, we also made 5 attempts to contact them by phone for the 6-month follow-up questionnaire. If after 5 attempts, we were still unable to contact a participant, he or she was considered lost to follow-up.

Definition of MTBI

All MTBIs were diagnosed by physicians in the ED. Operational criteria for clinical identification of MTBI included (1) mechanical energy to the head from external physical forces; (2) 1 or more of the following: confusion or disorientation, loss of consciousness for 30 minutes or less, or posttraumatic amnesia for less than 24 hours, and 1 or more of the following

transient neurological abnormalities: seizure, dizziness or unsteadiness, memory difficulty or forgetfulness, poor concentration or attention, vision problems, sleep disturbance, unusual fatigue, or tiredness; and (3) Glasgow Coma Scale score of 13 to 15 at 30 minutes postinjury or later upon presentation for healthcare. These manifestations of MTBI could not be due to drugs, alcohol, medications, caused by other injuries or treatment of other injuries (eg, systemic injuries, facial injuries, or intubation), caused by other problems (eg, psychological trauma, language barrier, or coexisting medical conditions), or caused by penetrating craniocerebral injury.²⁷ If a head computed tomographic (CT) scan revealed characteristics indicative of significant bleeding in the brain (eg, large subdural hematoma, epidural hematoma, or subarachnoid hemorrhage), the individual was excluded, because these abnormalities indicate a more severe traumatic brain injury. Individuals were included if CT abnormalities were less severe (eg, smear subdural hematoma, contusion(s), or edema).

Recovery outcome measures

Given the dynamic meaning of recovery for participants,²⁸ we used 4 measures of recovery, all of which were collected at both baseline and 6-month follow-up. The 4 measures used were as follows:

1. *Postconcussion symptoms (PCSs)*. Postconcussion symptoms were measured with the Rivermead Postconcussion Questionnaire (RPQ), a validated¹⁰ checklist assessing 16 common symptoms rated on a 5-point Likert scale. Participants were asked to rate how problematic, if at all, each symptom was experienced compared with their condition before their brain injury.²⁹ Although the RPQ is often used to measure PCS, a gold standard for classification of "mild" versus "severe" PCS is lacking. We defined an unfavorable outcome as a score of 2 (mild problem, but not interfering with daily activities), 3 (a moderate problem), or 4 (a severe problem) on at least 3 of the 16 symptoms. This definition has been previously used by a research team that also examined the internal construct validity of the RPQ.^{30,31}
2. *Glasgow Outcome Scale—Extended version (GOSE)*. The GOSE is a practical index of social outcome following head injury.^{32,33} It is a simple, hierarchical rating scale with 7 broad categories. The scale focuses on how head injury affects function in major life areas and is not intended to provide detailed information on specific deficits.³⁴ Participants rate themselves as severely disabled (unable to support themselves for 24 hours in society), moderately disabled (significant restrictions in lifestyle or work capacity, or both), or having good recovery (have resumed their previous lifestyle).³⁵ The participants' overall rating is based on the poorest outcome category indicated on the scale. The GOSE has shown adequate test-retest,³⁶ interobserver reliability,³³ excellent validity,³⁷ and good responsiveness.³⁸
3. *SF-12*. The SF-12 is a multipurpose short form survey with 12 questions selected from the SF-36 Health Survey.³⁹ The questions were combined, scored, and weighted to create 2 scales that provide glimpses into mental and physical functioning and overall health-related quality of life. The SF-12 is a generic measure and does not target specific age or disease groups. It has been developed to provide a shorter, yet valid alternative to the SF-36, which many health researchers consider too long to administer in studies with large samples. The SF-12 is weighted and summed to provide easily interpretable scales for physical and mental health. Physical and Mental Health Composite Scores (PCS and MCS) are computed using the scores of 12 questions and range from 0 to 100, where a 0 score indicates the lowest level of health measured by the scales and 100 indicates the highest level of health. The SF-12 has been tested and validated in persons with traumatic brain injury.⁴⁰
4. *Self-assessment of recovery*. Participants' perception of their own recovery was measured by asking: "How well do you feel you are recovering from your injuries?" Participants who responded "completely better" or "much improved" were considered recovered. Other response options included "some improvement", "no improvement", "getting a little worse", and "getting much worse". The global perceived recovery question is a reliable, valid, and responsive measure of health status.⁴¹ The ordinal transition scale has been used in previous randomized trials and cohort studies of neck pain.⁴²⁻⁴⁴ Subjects who report having "recovered" report significantly lower levels of pain and disability than those who are not recovered.

Prognostic markers

Potential prognostic marker information was collected from both clinical chart data and self-reported measures from participants at baseline and at 6-month follow-up.

Clinical data

Clinical data were extracted from the ED chart for participants who provided verbal consent at the time of the baseline interview. The obtained information included date of injury, date of visit to ED (if different from date

of injury), location where injury occurred (eg, home), activity during injury, mechanism of injury, Glasgow Coma Scale, posttraumatic amnesia duration, loss of consciousness duration, CT characteristics, early symptoms (such as dizziness, nausea/vomiting, headache), medication use, and physician rating of global recovery. Other concurrent injuries were also recorded.

Baseline measures

Self-reported demographic information collected included age, gender, education, income, height, weight, ethnicity, and marital status. We also assessed preinjury health and prior history of brain injury through self-report. Participants were also asked to rate their expectation of recovery by answering the following question: “Do you think that your injury will . . .” (1) get better soon; (2) get better slowly; (3) never get better; (4) I’m already better; or (5) don’t know.

The presence of comorbidities was measured with the Saskatchewan Comorbidity Scale, a patient-centered, self-report measure of health problems (eg, arthritis, diabetes). The 12 items ask the patient to indicate whether they experience a particular health problem, and if so, what effect that problem has on their health (none, mild, moderate, or severe). It has been demonstrated to have good test-retest reliability over a 3-week period and adequate validity.

Baseline and 6-month follow-up measures

The following were self-reported at both the baseline and 6-month interview:

Postinjury treatment/rehabilitation: We asked which types of healthcare providers were consulted (physician, chiropractor, physiotherapist, massage therapist).

Litigation seeking: Involvement of a lawyer was measured with 2 questions: “Have you consulted or hired a lawyer/paralegal to help you seek compensation for your injury?” and if so, “Is your lawyer supporting/helping you to quickly recover?”

Pain intensity: Pain intensity was measured with a Numerical Rating Scale for each of the following anatomical locations: (1) neck; (2) shoulder; (3) low back; (4) headache; (5) arms; (6) hands; (7) legs; (8) face; (9) feet; (10) midback; and (11) abdomen, chest, or groin. Participants were asked to rate their average pain from 0 to 10, where 0 means no pain at all and 10 means pain as bad as it could be. The Numerical Rating Scale has good validity and reliability in various clinical populations.^{45,46}

Depression: Depression was measured with the Centre for Epidemiological Studies–Depression Scale. The Centre for Epidemiological Studies–Depression Scale has high test-retest reliability, good internal consistency, and good factorial and discriminant validity.^{47–51} The Centre for Epidemiological Studies–Depression Scale

is scored from 0 to 60, with higher scores suggesting more depressive symptoms.

Posttraumatic stress: The Impact of Event Scale–Revised was used to measure the subjective response to a specific traumatic event. It is validated and appropriate for use in the senior population.⁵² It includes response sets of intrusion (intrusive thoughts, nightmares, intrusive feelings and imagery, dissociative-like reexperiencing), avoidance (numbing of responsiveness, avoidance of feelings, situations, and ideas), and hyperarousal (anger, irritability, hypervigilance, difficulty concentrating, heightened startle), as well as a total subjective stress Impact of Event Scale–Revised score. Scores range from 0 to 88, with higher scores indicating increased stress.⁵³

Fatigue: Fatigue was measured with the Fatigue Severity Scale.⁵⁴ Internal consistency is high for specific illness groups (MS and lupus) and healthy controls. In a study that compared the Fatigue Severity Scale with the Fatigue Scale, both were found to be useful measures of fatigue-related symptomatology within a general population of individuals with varying levels of fatigue.⁵⁵ However, the Fatigue Severity Scale appeared to represent a more accurate and comprehensive measure of fatigue severity and functional disability for individuals with chronic fatigue syndrome-like symptomatology. In addition, the Fatigue Severity Scale is a more practical measure due to its brevity and ease of administration and scoring. Higher scores indicate higher fatigue levels.

Coping: The abbreviated version of the Vanderbilt Pain Management Inventory was used to measure the ways in which individuals cope with pain.^{56,57} The measure has 2 subscales, an Active Coping scale with items such as “keeping busy or active” and a Passive Coping scale, with items such as “taking medication for purposes of immediate pain relief.” Patients rate the frequency with which they engage in these behaviors when in moderate pain, using a 5-point verbal rating scale. Higher scores reflect greater frequency of engaging in a particular type of coping strategy. The Vanderbilt Pain Management Inventory has acceptable psychometric properties.^{56–58}

Data analysis

All variables were checked for errors and inconsistencies. We compared baseline data between those lost to follow-up and those followed at 6 months in tabular format to assess the likelihood of attrition bias. Means or proportions were estimated as appropriate with associated standard deviations or 95% confidence intervals. Recovery measures were compared at baseline and at 6-month follow-up with McNemar test for proportions and paired *t* tests for means.

Bivariate associations were determined using contingency tables and χ^2 tests between baseline predictors

and unfavorable outcome on the RPQ, GOSE, and self-assessment of recovery at 6 months. Risk ratios and 95% confidence intervals were reported. Bivariate associations were assessed using *t* tests between baseline predictors and health scores on the mental and physical components of the SF-12 at 6 months. Mean differences in exposure groups were compared using 1-way analysis of variance. The Bonferroni adjustment was used to account for multiple comparisons. Individual predictors for each outcome measure were assessed separately. All statistical analyses were conducted using Stata, version 13.0.

RESULTS

Demographics

Fifty-four of 97 eligible individuals agreed to participate in the study (55.7% participation rate). Three participants who sustained a second head or brain injury during the follow-up period at 3, 12, and 18 weeks, respectively, had more unfavorable outcomes at both baseline and follow-up (data not shown). Because the second injury could complicate interpretation of our data with respect to the dependent variable of primary interest, these 3 individuals were removed from further analyses. Participants were predominantly white, female, and married, with a mean age of 76.2 years (SD: 7.4) (see Supplemental Digital Content Table 1, available at: <http://links.lww.com/JHTR/A167>, which presents baseline sample demographics). Eligible nonparticipants were also more likely to be female and had an average age (mean age: 78.5 y [SD: 8.1]) similar to that of participants but a higher proportion of adults older than 85 years. Forty-six of the 51 were followed up at 6 months (90.2% follow-up rate). Those lost to follow-up at 6 months were older (mean age 85.6 [SD: 9.3]) than those completing the study.

Clinical prognostic markers

Supplemental Digital Content Table 2, available at: <http://links.lww.com/JHTR/A168>, presents baseline clinical data recorded from the ED chart for those who permitted access. All participants allowed access and 17 nonparticipants allowed chart access as well. There were no differences between the clinical chart data of nonparticipants and participants (data not shown). Those lost to follow-up took longer to arrive at the ED after their injury, obtained more CT scans, had less nausea/vomiting, and more concurrent injuries (data not shown). One participant and 1 nonparticipant had positive CT findings considered thin smear subdural hematomas with no parenchymal hemorrhage, hydrocephalus, or herniation. No intervention was provided, and these patients

were not admitted to hospital but observed as outpatients.

Self-reported prognostic markers

At baseline, most participants reported low pain intensity, high fatigue, and same or better health since the injury (see Supplemental Digital Content Table 3 available at: <http://links.lww.com/JHTR/A169>, which presents baseline self-reported characteristics). Those lost to follow-up at 6 months tended to report worse health prior to the injury but good health since the injury compared with those fully participating in the study (data not shown). Those lost to follow-up were also more likely to indicate a good self-prognosis, have fewer painful body areas, have higher posttraumatic stress, use more passive coping strategies, and have more comorbidities but with a smaller impact than full participants.

Recovery at baseline and 6-month follow-up

The percentage of participants classified as not recovered varied across the different measures used to assess recovery at both baseline and 6-month follow-up. All recovery measures reflected improved recovery over the 6-month follow-up period, but the most notable improvements were detected with the global perceived recovery (self-report) (79.6%; 95% confidence interval [CI]: 65.5-88.9 not recovered at baseline and 26.5%; 95% CI: 15.8-41.1 at 6-month follow-up; $P < .0001$) and life-functioning (GOSE) measures (59.2%; 95% CI: 44.5-72.4 not recovered at baseline, and 20.4%; 95% CI: 11.1-34.5 at 6-month follow-up; $P < .0001$). Post-concussion symptoms (RPQ) improved from baseline (38.8%; 95% CI: 25.9-53.5) compared with 6 months later (14.3%; 95% CI: 6.8-27.6), $P = .0009$. Improvement in SF-12 mental health functioning (71.5; SD: 21.1-84.1; SD: 15.4; $P = .0001$) and physical health functioning (46.9; SD: 28.9-74.3; SD: 25.0; $P < .0001$) was also apparent. Those lost to follow-up were more likely to show better recovery at baseline on all measures except for global perceived recovery, where the proportion reporting recovery was approximately 20% for both full participants and those lost to follow-up (data not shown).

Prognostic markers and poor recovery

Table 1 highlights potentially important prognostic markers of poor recovery for the 3 categorical measures of recovery. The only baseline factor independently associated with all 3 measures of recovery was worse health 1 year prior to injury. Poor self-prognosis and baseline depression were associated with high-risk ratios across the 3 measures but did not reach statistical significance. Those reporting high fatigue at baseline

TABLE 1 Risk ratios for the independent associations between baseline markers and poor recovery 6 months later in older adults visiting the emergency department for a mild traumatic brain injury according to varying measures of recovery

Baseline markers	Global perceived recovery (self-report) RR (95% CI)	Postconcussion symptoms (RPQ) RR (95% CI)	Life functioning (GOSE) RR (95% CI)
Location of injury			
Not at home	1.00	1.00	1.00
Home	0.64 (0.20-2.00)	0.80 (0.13-5.07)	0.80 (0.19-3.44)
Environment of injury			
Outdoors	1.00	1.00	1.00
Indoors	0.50 (0.15-1.66)	1.16 (0.18-7.45)	0.87 (0.22-3.40)
Mechanism of injury			
Other	1.00	1.00	1.00
Fall	0.31 (0.12-0.84)	0.28 (0.04-1.78)	0.42 (0.07-2.44)
Posttraumatic amnesia	0.69 (0.10-4.85)	2.07 (0.22-19.73)	1.04 (0.14-7.89)
Loss of consciousness	2.48 (0.65-9.48)	...	6.20 (1.11-34.53)
Dizziness	3.62 (1.25-10.47)	3.10 (0.58-16.63)	1.55 (0.40-6.06)
Nausea/vomiting	2.36 (0.88-6.34)	1.89 (0.36-9.97)	1.13 (0.25-5.09)
Headache	1.35 (0.46-3.97)	1.15 (0.21-6.26)	4.62 (0.60-35.32)
Other symptoms	1.71 (0.61-4.78)	5.68 (0.69-46.94)	3.55 (0.77-16.43)
Concurrent injuries	1.69 (0.42-6.71)	1.50 (0.19-12.11)	...
Health 1 y before injury			
Much/somewhat better or same	1.00	1.00	1.00
Somewhat or much worse	2.71 (1.04-7.11)	7.13 (1.41-35.95)	6.33 (1.75-22.97)
Health month before injury			
Excellent or very good	1.00	1.00	1.00
Good or fair	0.94 (0.32-2.75)	2.47 (0.46-13.32)	1.24 (0.31-4.86)
Health since injury			
Excellent or very good	1.00	1.00	1.00
Good or fair	1.93 (0.58-6.45)	...	1.65 (0.34-8.06)
Poor	1.58 (0.22-11.58)	...	2.38 (0.28-20.28)
Proportion reporting worse health since injury	1.90 (0.70-5.21)	3.43 (0.64-18.31)	1.71 (0.44-6.67)
Past head/brain injury	1.90 (0.70-5.21)	1.52 (0.29-8.14)	3.05 (0.78-11.86)
Self-prognosis			
Already cured/get better soon	1.00	1.00	1.00
Get better slowly or never (poor prognosis)	2.29 (0.86-6.14)	4.13 (0.78-21.75)	2.06 (0.54-7.90)
Healthcare providers consulted			
Any healthcare provider	2.09 (0.73-6.00)	1.39 (0.18-10.69)	2.23 (0.53-9.30)
Physician	1.82 (0.54-6.17)	2.05 (0.28-14.92)	1.37 (0.20-9.16)
Other healthcare provider (not including chiropractor, physiotherapist, or massage therapist)	4.89 (2.73-8.76)	0	3.67 (0.76-17.67)
Number of body areas in pain			
1 to <6	1.00	1.00	1.00
6 to <11	1.36 (0.48-3.87)	1.59 (0.30-8.43)	0.95 (0.21-4.30)
High pain intensity ^a	0.75 (0.26-2.20)	0.88 (0.16-4.74)	0.99 (0.25-3.90)
Depression ^b	1.82 (0.54-6.17)	2.05 (0.28-14.92)	1.37 (0.20-9.16)
High posttraumatic stress ^c	0	0	0
High fatigue ^d	4.13 (1.00-17.04)	3.67 (0.44-30.35)	...
Coping ^e			
Low passive	1.00	1.00	1.00
High passive	2.38 (0.87-6.52)	9.50 (1.98-45.59)	2.53 (0.63-10.23)

(continues)

TABLE 1 Risk ratios for the independent associations between baseline markers and poor recovery 6 months later in older adults visiting the emergency department for a mild traumatic brain injury according to varying measures of recovery (Continued)

Baseline markers	Global perceived recovery (self-report) RR (95% CI)	Postconcussion symptoms (RPQ) RR (95% CI)	Life functioning (GOSE) RR (95% CI)
Number of comorbidities ^f			
1 to < 6	1.00	1.00	1.00
6 to ≤ 12	0.69 (0.17-2.79)	1.83 (0.35-9.67)	0.46 (0.06-3.43)
Large impact of comorbidity on health ^f	2.21 (0.76-6.44)	3.32 (0.62-17.72)	2.95 (0.76-11.47)
Anticoagulant medication use	0.38 (0.11-1.24)	0.25 (0.03-2.07)	0.40 (0.09-1.86)
Pain medication use	1.88 (0.57-6.17)	1.06 (0.19-5.72)	0.94 (0.24-3.72)
New medication since injury	3.28 (1.13-9.55)	1.25 (0.23-6.73)	2.50 (0.64-9.82)
LOC (self-reported)	3.10 (1.12-8.55)	2.07 (0.40-10.67)	4.13 (1.11-15.42)
PTA (self-reported)	4.81 (1.71-13.56)	4.13 (0.78-21.75)	6.88 (1.53-30.84)
PTA duration (self-reported)			
<1 h	1.00	1.00	1.00
>1 h	...	...	...
Disorientation/confusion (self-reported)	2.72 (0.93-7.99)	2.33 (0.43-12.62)	2.07 (0.52-8.20)
Disorientation/confusion duration (self-reported)			
<1 h	1.00	1.00	1.00
>1 h	0.83 (0.15-4.64)	2.50 (0.32-19.53)	1.67 (0.25-11.07)
Number of postconcussion symptoms (self-reported)			
1 to <7	1.00	1.00	1.00
7 to ≤14	2.35 (0.89-6.18)	4.23 (0.81-22.05)	3.76 (0.99-14.20)
Postconcussion symptoms (self-reported)			
Feeling of numbness, tingling, or pain in arms or hands	2.33 (0.75-7.28)	2.63 (0.38-18.20)	4.20 (1.17-15.10)
Feeling of numbness, tingling, or pain in legs or feet	0.82 (0.13-5.13)	0	1.37 (0.20-9.16)
Memory problems/forgetfulness	2.06 (0.75-5.64)	2.40 (0.46-12.45)	4.80 (1.28-18.02)
Concentration/attention problems	3.62 (1.25-10.47)	3.10 (0.58-16.63)	5.17 (1.13-23.61)
Irritability	1.82 (0.65-5.07)	2.12 (0.40-11.11)	1.27 (0.29-5.67)
Vision problems	3.05 (1.12-8.27)	1.69 (0.32-8.99)	3.38 (0.88-13.09)
Hearing problems	3.19 (1.18-8.56)	9.56 (2.47-36.98)	2.39 (0.41-13.94)
Sleep problems	1.07 (0.37-3.12)	1.25 (0.23-6.73)	0.75 (0.16-3.44)
Unusual fatigue or tiredness	1.88 (0.57-6.17)	2.81 (0.34-23.25)	1.76 (0.38-8.13)
Anxiety/worry	1.56 (0.55-4.39)	...	1.73 (0.44-6.88)
Pain when neck is moved	0.54 (0.19-1.50)	0.16 (0.02-1.33)	0.26 (0.06-1.19)
Reduced ability to move neck	1.72 (0.62-4.75)	1.38 (0.26-7.39)	1.55 (0.40-6.06)
Sore jaw	2.71 (1.04-7.11)	3.17 (0.63-15.98)	3.56 (0.98-12.92)

Abbreviations: CI, confidence interval; GOSE, Glasgow Outcomes Scale—Extended; LOC, loss of consciousness; PTA, posttraumatic amnesia; RPQ, Rivermead Postconcussion symptom Questionnaire; RR.

^aNumerical Rating Scale.

^bCES-D.

^cImpact of Event Scale (Revised).

^dFatigue Severity Scale.

^ePain Management Inventory.

^fSaskatchewan Comorbidity Scale.

were 4 times more likely to self-report poor recovery. Self-reported posttraumatic amnesia at baseline was associated with a 4.81 times increased risk of self-reported poor recovery and a 6.88 times increased risk of poor life functioning 6 months later. Similarly, concentration and attention problems were associated with a 3.62 times increased risk of self-reported nonrecovery and a 5 times increased risk of poor life functioning 6 months later. Those reporting hearing problems at baseline were 3 times more likely to self-report poor recovery and 9 times more likely to have worse PCSs than those not reporting hearing problems. Self-reported loss of consciousness and posttraumatic amnesia at baseline were both associated with worse self-reported recovery and life functioning.

The only baseline factors associated with worse physical and mental functioning at 6 months were worse health 1 year before the injury (see Supplemental Digital Content Table 4, available at: <http://links.lww.com/JHTR/A170>, which presents mental and physical functioning at 6 months). Other markers associated with reduced physical health functioning included poor health 1 month before the injury, poor health since injury, and a large impact of comorbidity on health. Other markers associated with reduced mental health functioning included high fatigue, hearing problems, and loss of consciousness as a symptom reported on the clinical chart.

DISCUSSION

We assessed factors reported shortly after MTBI in older adults for their prognostic influences on 5 measures of recovery 6 months later. Those reporting worse health 1 year before the injury, poor expectations for recovery, depression, and fatigue shortly after their injury reported worse recovery at 6 months according to most recovery measures used in this study.

Our findings suggest that recovery after MTBI in older adults may be more associated with psychosocial factors than with biomedical or injury-related phenomena. This finding is similar to that suggested for prognosis in younger adults. A recent systematic review of self-reported prognosis in adults younger than 65 years after MTBI highlighted the importance of several psychosocial determinants of recovery such as poor expectations for recovery and negative injury perceptions.^{3,26} We found poorer premorbid health status, poor expectations for recovery, depressive symptomatology, and fatigue to be prognostic markers of recovery in older adults.

Stulemeijer et al¹⁰ found that the absence of comorbid physical problems predicted good recovery according to the RPQ in MTBI patients between the ages of 18 and 60 years admitted to the ED of a level I trauma center. We found that older adults reporting worse health

in the year prior to the injury reported worse recovery on the RPQ, a self-report index of recovery, life functioning (GOSE), and physical and mental health functioning (SF-12). Poor premorbid health may leave MTBI sufferers in inadequate physical shape to overcome the additional strain of an MTBI regardless of age. Alternatively, symptoms related primarily to preexisting problems or comorbidity may be inaccurately attributed to the MTBI.^{26,59}

Psychosocial factors such as poor expectations for recovery have been suggested to be associated with recovery after MTBI in adult populations.⁶⁰ In a prospective, population-based, inception cohort study of individuals with MTBI after motor vehicle collisions, Cassidy et al⁶⁰ found that, within 42 days of their collision, those expecting to never get better were 74% less likely to self-report recovery over a 1-year period (hazard rate ratio [HRR] = 0.26; 95% CI: 0.14–0.50). We found that older adults who expected to get better slowly or never get better were 229% (risk ratio [RR] = 2.29; 95% CI: 0.86–6.14) more likely to report poor recovery at 6 months postinjury. We also found poor recovery expectations in older adults to be associated with a 4 times increased risk for reporting PCSs and a 2 times increased risk for reporting a reduction in life functioning 6 months later. We found less of an effect size between recovery expectations and physical or mental health functioning. It is important to note that our estimates were not statistically significant, but this is likely due to statistical power limitations associated with the pilot study design.

Depression has also been shown to be a weak risk factor for delayed recovery from MTBI in the adult population. Cassidy et al found that for every 1 unit reported increase in depression on the Centre for Epidemiologic Studies Depression scale, there was an associated 1% (HRR = 0.99; 95% CI: 0.99–1.00) decrease in recovery over the 1-year postinjury period.⁶⁰ Gil et al⁶¹ recruited 120 MTBI patients from 2 surgical wards of a hospital in Israel and found that those with any depressive symptoms within the week after injury (as indicated by the Beck Depression Inventory) were 5 times more likely to have posttraumatic stress disorder (odds ratio = 5.1, 95% CI: 1.0–9.2) 6 months later. Norrie et al⁶² found those with generalized depression 3 months after their ED visit for an MTBI were more likely to have fatigue at 6 months. We found depression to have a strong effect on mental health functioning 6 months later in the older adult population. We did not find a statistically significant association with perceived recovery, PCSs, life functioning, or physical health functioning, but again this may have been due to the small sample size under study.

We found high fatigue to be associated with perceived recovery and mental health functioning. Fatigue is often

reported after MTBI and may be associated with higher levels of baseline psychosocial factors.⁶³ A larger study is needed with the ability to control for confounding factors to determine whether fatigue is an independent predictor of recovery after MTBI in older adults or whether it is only highly correlated with psychosocial prognostic markers.

A few case series and descriptive studies have suggested that elderly people have poorer recovery after MTBI.^{11–13} Although we did not directly compare recovery in older adults with that of younger adult MTBI patients, we found proportions of nonrecovery at 6 months ranging from 10.9% still experiencing PCSs to 23.9% self-reporting nonrecovery. These estimates are no greater than the 22% to 36% of adult MTBI patients who continue to report 3 or more posttraumatic symptoms 6 months after injury.²⁶

To our knowledge, this is the first cohort study examining prognostic markers for MTBI in older adults. Goldstein et al⁴ studied 35 elderly TBI patients (aged ≥ 50 years; 18 MTBI and 17 moderate TBI) prospectively recruited by acute care neurosurgery services at 2 university hospital medical schools in the United States. They compared cognitive and neurobehavioral measures up to 2 months postinjury and found that survivors of moderate TBI performed significantly poorer than the MTBI patients on most cognitive measures. Both moderate and MTBI patients exhibited greater depression and anxiety than a group of uninjured controls. We found that older patients reporting depression after MTBI were more likely to have prolonged recovery in mental health functioning.

Study limitations

Our study has several limitations. First, since this was a pilot study, the small sample size limited the statistical power and provided no control for confounding factors. Therefore, this study can offer, at best, phase I evidence for potential prognostic markers of recovery after MTBI.^{3,14,26} Second, with a 55% participation rate, we cannot rule out selection bias or ensure that our sample is generalizable to the entire older adult MTBI population. However, we were able to compare limited demographic information in nonparticipants and participants, and this suggested that our study sample may have fewer MTBI patients older than 85 years than the total MTBI population. Therefore, our findings may be more generalizable to those aged 65 to 84 years. Future studies of MTBI in the older adult population should account for the difficulties in recruiting the oldest old by offering face-to-face recruitment strategies, ensuring the length of the survey is no longer than 15 to 20 minutes to account for fatigue, and providing an extended period for recruitment as enlisting older adults

may take more than 1 meeting.⁶⁴ Third, although all MTBIs were diagnosed in the ED by physicians using the World Health Organization task force recommendations, the challenge remains to differentiate symptoms of concussion from comorbid symptoms that may exist in older adult patients. Fourth, the accuracy and completeness of clinical information in ED charts may lead to misclassification. However, this would likely lead to conservative estimates, as it is unlikely that misclassification would differ between those with delayed and quick recovery. Similarly, misclassification could occur through participants recalling preinjury states, or malingering or denying their symptoms. Again, the prospective nature of the study design would lead this misclassification to likely be nondifferential in nature, resulting in conservative estimates of association. Fifth, although our attrition rate was low (9.8%), there is still a possibility of attrition bias.⁶⁵ Small numbers made it difficult to compare those lost to follow-up to full participants. However, it appeared that those lost to follow-up were more likely to be older, have more concurrent injuries and comorbidities, have greater posttraumatic stress, and use more passive coping strategies. Yet, they tended to have a good expectation for recovery and score higher on all recovery measures at baseline except for global perceived recovery. Therefore, it is unlikely that attrition in this group has produced significantly biased estimates. Finally, one research assistant conducted the baseline assessment and the 6-month follow-up. This provides the opportunity for information bias; however, we felt that this was unlikely given (1) the 6-month interval; (2) that the outcome scores were not calculated until after all follow-up data were collected; and (3) that all baseline data were stored separately from follow-up data such that the baseline data were not available to the research assistant at the time of the follow-up survey.

CONCLUSION

Mild traumatic brain injury results in significant disability for older adults. A considerable number of older adults continue to struggle with recovery at least 6 months after their injury. Recovery after MTBI in older adults may be more associated with psychosocial factors rather than biomedical or injury-related factors. Using 5 measures of recovery, we found that those reporting worse health 1 year before the injury, poor expectations for recovery, depression, and fatigue shortly after their injury reported worse outcomes 6 months later.

This pilot study has generated 3 hypotheses that require further testing. First, we need to confirm these prognostic markers in a large prospective cohort study. Second, findings from this pilot study suggest that, contrary to previous reports, older adults may not have

poorer recovery after MTBI than younger adults. This needs to be confirmed in a study designed to specifically test this hypothesis. Third, one purpose of prognosis is the early recognition of patients at risk of a poor outcome. Clinical prediction tools can inform clinicians

and patients of the likely course of recovery and help clinicians make treatment decisions about their elderly patients with MTBI. These pilot findings provide preliminary evidence for the development of a clinical prediction tool in the older adult MTBI population.

REFERENCES

1. Cassidy JD, Carroll LJ, Peloso PM, et al. Incidence, risk factors and prevention of mild traumatic brain injury: results of the WHO Collaborating Centre Task Force on Mild Traumatic Brain Injury. *J Rehabil Med Suppl.* 2004;43:28–60.
2. Carroll LJ, Cassidy JD, Peloso PM, et al. Prognosis for mild traumatic brain injury: results of the WHO Collaborating Centre Task Force on Mild Traumatic Brain Injury. *J Rehabil Med Suppl.* 2004;43:84–105.
3. Cassidy JD, Cancelliere C, Carroll LJ, et al. Systematic review of self-reported prognosis in adults after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S132–S151.
4. Goldstein FC, Levin HS, Goldman WP, et al. Cognitive and neurobehavioural functioning after mild versus moderate traumatic brain injury in older adults. *J Int Neuropsychol Soc.* 2001;7(3):373–383.
5. Karasik D, Demissie S, Cupples LA, Kiel DP. Disentangling the genetic determinants of human aging: biological age as an alternative to the use of survival measures. *J Gerontol A Biol Sci Med Sci.* 2005;60:574–587.
6. McDonald SWS, DeCarlo CA, Dixon RA. Linking biological and cognitive aging: towards improving characterizations of developmental time. *J Gerontol B Psychol Sci Soc Sci.* 2011;66:59–70.
7. Thompson HJ, McCormick WC, Kegan SH. Traumatic brain injury in older adults: epidemiology, outcomes, and future implications. *J Am Geriatr Soc.* 2008;54:1590–1595.
8. Anderson AJ. Treatment of depression in older adults. *Int J Psychosoc Rehabil.* 2002;6:69–78.
9. Riedel-Heller SG, Busse A, Angermeyer MC. The state of mental health in old-age across the ‘old’ European Union—a systematic review. *Acta Psychiatr Scand.* 2005;113:388–401.
10. Stulemeijer M, van der Werf S, Borm GF, Vos PE. Early prediction of favourable recovery 6 months after mild traumatic brain injury. *J Neurol Neurosurg Psych.* 2008;79:936–942.
11. Amacher AL, Bybee DE. Toleration of head injury by the elderly. *Neurosurg.* 1987;20:954–958.
12. Pentland B, Jones PA, Roy CS, Miller JD. Head injury in the elderly. *Age Ageing.* 1986;15:193–202.
13. Ritchie PD, Cameron PA, Ugoni AM, Kaye AH. A study of the functional outcome and mortality in elderly patients with head injuries. *J Clin Neurosci.* 2000;7:301–304.
14. Salmi LR, Cassidy JD, Holm L, Cancelliere C, Côté P, Borg J. Introduction to the findings of the International Collaboration on Mild Traumatic Brain Injury prognosis: what is a prognostic study? *Arch Phys Med Rehabil.* 2014;95(3 suppl):S95–S100.
15. Cancelliere C, Cassidy JD, Li A, Donovan J, Côté P, Hincapié CA. Systematic search and review procedures: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S101–S131.
16. Carroll LJ, Cassidy JD, Cancelliere C, et al. Systematic review of the prognosis after mild traumatic brain injury in adults: cognitive, psychiatric, and mortality outcomes: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S152–S173.
17. Hung R, Carroll LJ, Cancelliere C, et al. Systematic review of the clinical course, natural history, and prognosis for pediatric mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S174–S191.
18. Keightley ML, Côté P, Rumney P, et al. Psychosocial consequences of mild traumatic brain injury in children: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S192–S200.
19. Cancelliere C, Kristman VL, Cassidy JD, et al. Systematic review of return to work after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S201–S209.
20. Cancelliere C, Hincapié CA, Keightley M, et al. Systematic review of prognosis and return to play after sport concussion: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S210–S229.
21. Boyle E, Cancelliere C, Hartvigsen J, Carroll LJ, Holm LW, Cassidy JD. Systematic review of prognosis after mild traumatic brain injury in the military: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S230–S237.
22. Marras C, Hincapié CA, Kristman VL, et al. Systematic review of the risk of Parkinson’s disease after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S238–S244.
23. Godbolt AK, Cancelliere C, Hincapié CA, et al. Systematic review of the risk of dementia and chronic cognitive impairment after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S245–S256.
24. Nygren-de Boussard C, Holm LW, Cancelliere C, et al. Nonsurgical interventions after mild traumatic brain injury: a systematic review. Results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S257–S264.
25. Kristman VL, Borg J, Godbolt AK, et al. Methodological issues and research recommendations for prognosis after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Arch Phys Med Rehabil.* 2014;95(3 suppl):S265–S277.
26. Donovan J, Cancelliere C, Cassidy JD. Summary of the findings of the International Collaboration on Mild Traumatic Brain Injury Prognosis. *Chiropractic Manual Ther.* 2014;22:38.
27. Carroll LJ, Cassidy JD, Holm L, Kraus J, Coronado VG. Methodological issues and research recommendations for mild traumatic brain injury: the WHO Collaborating Centre Task Force on Mild Traumatic Brain Injury. *J Rehabil Med Suppl.* 2004;43:113–125.
28. Beaton DE, Tarasuk V, Katz JN, Wright JG, Bombardier C. “Are you better?” A qualitative study of the meaning of recovery. *Arthritis Rheum.* 2001;45(3):270–279.
29. King N, Crawford S, Wenden F, Moss N, Wade D. Early prediction of persisting postconcussion symptoms following mild and moderate head injuries. *Br J Clin Psychol.* 1999;38:15–25.

30. Lannsjö M, Backheden M, Johansson U, Af Geijerstam JL, Borg J. Does head CT scan pathology predict outcome after mild traumatic brain injury? *Eur J Neurol*. 2013;20:124–129.
31. Lannsjö M, Borg J, Björklund G, Af Geijerstam JL, Lundgren-Nilsson A. Internal construct validity of the Rivermead Post-Concussion Symptoms Questionnaire. *J Rehabil Med*. 2011;43:997–1002.
32. Jennett B, Bond M. Assessment of outcome after severe brain damage. *Lancet*. 1975;1:480–484.
33. Jennett B, Snoek J, Bond MR, Brooks N. Disability after severe head injury: observations on the use of the Glasgow Outcome Scale. *J Neurol Neurosurg Psych*. 1981;44:285–293.
34. Wilson JTL, Pettigrew LEL, Teasdale GM. Structured interviews for the Glasgow Outcome Scale and the Extended Glasgow Outcome Scale: guidelines for their use. *J Neurotrauma*. 1998;15:573–585.
35. Wilson JTL, Pettigrew LEL, Teasdale GM. Emotional and cognitive consequences of head injury in relation to the Glasgow Outcome Scale. *J Neurol Neurosurg Psych*. 2000;69:204–209.
36. Maas AIR, Braakman R, Schouten HJA, Minderhoud JM, van Zomeren AH. Agreement between physicians on assessment of outcome following severe head injury. *J Neurosurg*. 1983;58:321–325.
37. Brooks DN, Hosie J, Bond MR, Jennett B, Aughton M. Cognitive sequelae of severe head injury in relation to the Glasgow Outcome Scale. *J Neurol Neurosurg Psych*. 1986;49:549–553.
38. Levin HS, Boake C, Song J, et al. Validity and sensitivity to change of the extended Glasgow Outcome Scale in mild to moderate traumatic brain injury. *J Neurotrauma*. 2001;18:575–584.
39. Ware J Jr, Kosinski M, Keller SD. A 12-Item Short-Form Health Survey: construction of scales and preliminary tests of reliability and validity. *Med Care*. 1996;34(3):220–233.
40. Gandek B, Ware JE, Aaronson NK, et al. Cross-validation of item selection and scoring for the SF-12 Health Survey in nine countries: results for the IQOLA Project. *J Clin Epidemiol*. 1998;51(11):1171–1178.
41. Fischer D, Stewart AL, Bloch DA, Lorig K, Laurent D, Holman H. Capturing the patient's view of change as a clinical outcome measure. *JAMA*. 1999;282:1157–1162.
42. Hoving JL, Koes BW, de Vet HC, et al. Manual therapy, physical therapy, or continued care by a general practitioner for patients with neck pain. A randomized, controlled trial. *Ann Intern Med*. 2002;21:136:713–722.
43. Hoving JL, de Vet HC, Koes BW, et al. Manual therapy, physical therapy, or continued care by the general practitioner for patients with neck pain: long-term results from a pragmatic randomized clinical trial. *Clin J Pain*. 2006;22:370–377.
44. Cassidy JD, Carroll LJ, Côté P, Frank J. Does multidisciplinary rehabilitation benefit whiplash recovery? Results of a population-based incidence cohort study. *Spine*. 2007;32:126–131.
45. Jensen MP, Karoly P, Braver S. The measurement of clinical pain intensity: a comparison of six methods. *Pain*. 1986;27:117–126.
46. Wevers ME, Lowe NK. A critical review of analogue scales in the measurement of clinical phenomena. *Res Nurs Health*. 1990;13:227–236.
47. Radloff LS. The CES-D Scale: a self-report depression scale for research in the general population. *Appl Psych Measure*. 1977;1(3):385–401.
48. Boyd JH, Weissman MM, Thompson WD, Myers JK. Screening for depression in a community sample. *Arch Gen Psych*. 1982;39:1195–1200.
49. Devins GM, Orme CM, Costello CG, et al. Measuring depressive symptoms in illness populations: psychometric properties of the Centre for Epidemiologic Studies Depression (CES-D) Scale. *Psych Health*. 1988;2:139–156.
50. Orme JG, Reis J, Herz EJ. Factorial and discriminant validity of the Center for Epidemiological Studies Depression (CES-D) Scale. *J Clin Psych*. 1986;42(1):28–33.
51. Zich JM, Attkisson CC, Greenfield TK. Screening for depression in primary care clinics: the CES-D and the BDI. *Int J Psych Med*. 1990;20(3):259–277.
52. Hartford Institute for Geriatric Nursing, New York university, College of Nursing. Best practice information on care of older adults. www.ConsultGeriRN.org 2008. Accessed February 1, 2016.
53. Weiss DS. The impact of event scale: revised. In: Wilson JP, Tang CS eds. *Cross-Cultural Assessment of Psychological Trauma and PTSD*. New York, NY: Springer; 2007:219–238.
54. Krupp LB, LaRocca NG, Steinberg AD. The Fatigue Severity Scale: application to patents with multiple sclerosis and systemic lupus erythematosus. *Arch Neurol*. 1989;46:1121–1123.
55. Taylor RR, Jason LA, Toarres A. Fatigue rating scales: an empirical comparison. *Psychol Med*. 2000;30:849–856.
56. Brown GK, Nicassio PM. Development of a questionnaire for the assessment of active and passive coping strategies in chronic pain patients. *Pain*. 1987;31:53–64.
57. Smith CA, Wallston KA, Dwyer KA. On babies and bathwater: disease impact and negative affectivity in the self-reports of persons with rheumatoid arthritis. *Health Psych*. 1995;14:64–73.
58. Brown GK, Nicassio PM, Wallston KA. Pain coping strategies and depression in rheumatoid arthritis. *J Consult Clin Psych*. 1989;57(5):652–657.
59. Mittenberg W, DiGiulio DV, Perrin S, et al. Symptoms following mild head injury: expectation as aetiology. *J Neurol Neurosurg Psychiatry*. 1992;55:200–204.
60. Cassidy JD, Boyle E, Carroll LJ. Population-based, inception cohort study of the incidence, course, and prognosis of mild traumatic brain injury after motor vehicle collisions. *Arch Phys Med Rehabil*. 2014;95(3 suppl):S278–S285.
61. Gil S, Caspi Y, Ben-Ari IZ, Koren D, Klein E. Does memory of a traumatic event increase the risk for posttraumatic stress disorder in patients with traumatic brain injury? A prospective study. *Am J Psychiatry*. 2005;162:963–969.
62. Norrie J, Heitger M, Leatham J, Anderson T, Jones R, Flett R. Mild traumatic brain injury and fatigue: a prospective longitudinal study. *Brain Inj*. 2010;24:1528–1538.
63. De Leon MB, Kirsch NL, Maio RF, et al. Baseline predictors of fatigue 1 year after mild head injury. *Arch Phys Med Rehabil*. 2009;90:956–965.
64. Jacelon CS. Older adults' participation in research. *Nurse Res*. 2007;14(4):65–73.
65. Kristman V, Manno M, Côté P. Loss to follow-up in cohort studies: how much is too much? *Eur J Epidemiol*. 2004;19(8):751–760.